Convective Organization as Markov Trajectories in Latent Space

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Abstract

Convective storms are stochastic, high-dimensional systems whose evolution resists compact description. Here we focus on their *internal* variability—the diversity of storm evolution that emerges under a fixed large-scale environment rather than from changes in the forcing itself. We introduce a framework that treats convective organization dynamically: an encoder maps isentropic mass-flux profiles into a low-dimensional latent space, the latent space is partitioned into discrete convective states, and transitions between states are modelled as a Markov chain. Applied to cloud-resolving simulations of squall lines under fixed vertical wind shear, the model identifies eight latent dimensions and seven states recovering the canonical phases of the convective life cycle. The learned dynamics expose two pathways: a direct route into sustained deep convection and a slower route through weakly organized phases. Departures from strict Markovianity in the organized regime expose the role of internal feedbacks—particularly cold-pool dynamics—as sources of effective memory; conditioning on cold-pool intensity at the starting time partly restores Markovian behaviour, suggesting that explicitly embedding such internal processes improves model fidelity. The results indicate that convective diversity arises from the range of dynamical trajectories supported by the environment rather than from distinct states alone, and offer a learned, stochastic alternative to prescribed multi-cloud parameterizations.

1 Introduction

Convective storms are high-dimensional dynamical systems: nonlinear couplings among thermodynamics, fluid motion and microphysics generate complex, multiscale and intrinsically stochastic evolution. Whether resolved by cloud-resolving simulations (Feng et al., 2025) or described by satellite observations (Hu, Leung, and Feng, 2020; Fiolleau and Roca, 2024), this complexity has so far resisted any compact, dynamically faithful representation. And yet convection exerts an outsized control on vertical transports of mass, energy and moisture, so its representation in coarse-grained climate models remains a persistent challenge—particularly under warming (Yang et al., 2023; Bolot et al., 2025; Prein et al., 2022; Chakraborty, Sullivan, and Feng, 2023). A further difficulty is that the convective response to a *fixed* large-scale forcing is not a single outcome but a broad distribution of storm evolutions (Abramian, C. Muller, Risi, et al., 2025); this *internal* variability, intrinsic to the convection rather than inherited from changes in the environment, is the focus of the present work.

This complexity has historically motivated drastic simplifications. Stochastic multi-cloud models

(Khouider and Majda, 2006; Majda and Stechmann, 2008; Khouider, Biello, and Majda, 2010) approximate convective populations as a Markov chain over a small number of prescribed cloud types—shallow, congestus and deep—with transition rules on a coarse lattice. Such models capture the birth, growth and decay of storm populations as well as the horizontal variability of cloud cover. Because the states are chosen *a priori* rather than learned from data, however, they cannot resolve the diversity of vertical mass-flux profiles that arise for a given cloud geometry. This limitation is most consequential during extreme events (C. J. Muller and Takayabu, 2020), where the dynamical component governs precipitation change (Singh and O'Gorman, 2012; Abramian, C. Muller, and Risi, 2023; Wang et al., 2022): whether a system is bottom- or top-heavy strongly influences the duration and intensity of extreme rainfall (Li et al., 2025).

More generally, low-dimensional descriptions of high-dimensional systems are often weakly connected to the underlying dynamics, because the variables chosen to describe the flow are not necessarily the ones in which the dynamics simplify. Koopman operator theory (Koopman, 1931; Brunton et al., 2021) offers a principled resolution. Rather than projecting the dynamics directly onto a small set of physical variables—where nonlinear couplings make any truncation dynamically inconsistent—one constructs *observables*: nonlinear transformations of the physical state in which the dynamics become exactly linear. The nonlinear complexity is not eliminated; it is transferred to the geometry of the observable space, so that the projected evolution is linear by construction regardless of how complex the underlying system is. In practice, the right observables must be approximated from data, a problem recently recast as representation learning (Lusch, Kutz, and Brunton, 2018): an encoder maps the physical state into a latent space where the dynamics are constrained to be linear, while a decoder ensures fidelity to the original state. Recent extensions cover chaotic and stochastic dynamics as well (Nathaniel and Gentine, 2026).

We build on this framework and make two specific choices that distinguish our approach. First, we cast the convective flow in isentropic coordinates (Pauluis and Mrowiec, 2013), which reorganize mass flux according to moist static energy. This yields a thermodynamically consistent representation that preserves energetic structure and filters reversible adiabatic motions. The isentropic field is itself high-dimensional—it is not a reduced description—but it provides a physically grounded substrate from which dynamically relevant observables can be extracted by the encoder.

Second, we depart from Koopman autoencoders that enforce *deterministic* linear dynamics. Instead, we partition the learned latent space into a finite set of discrete states through joint clustering (Kaiser et al., 2014; Souza, 2024) and model transitions between them probabilistically. The reduced-order model that emerges is a Markov chain on state occupation probabilities—finite-dimensional, linear, and memoryless by construction. Unlike existing stochastic convection models, the discrete states are neither prescribed nor heuristic: they are learned jointly with the encoder, so that the partition is shaped by the very dynamics it must represent.

The construction bridges two traditions. Stochastic multi-cloud models operate on prescribed states with limited dynamical diversity; Koopman autoencoders learn expressive representations but typically enforce deterministic dynamics. Combining *learned* discrete states with *stochastic* transitions yields a model that is at once physically interpretable and consistent with the irreducibly stochastic character of convective evolution.

We apply this framework to squall lines simulated with the System for Atmospheric Modeling (SAM) under controlled large-scale forcing and address three questions:

(i) What are the dominant convective mass-flux states accessible under a given forcing, here characterized by vertical wind shear?

(ii) Can transitions between these states be approximated by a Markov chain, and do the pathways differ between weakly and strongly organized convection?

(iii) To what extent is convective variability memoryless, and where do internal feedbacks—cold-pool dynamics in particular—introduce effective memory?

The remainder of the paper is organized as follows. Section 2 describes the methodology. Section 3 presents the results. Section 4 discusses implications for convective modelling and concludes.

2 Methods

2.1 Idealized cloud-resolving simulations

To examine the response of convection to fixed vertical wind shear, we perform idealized cloud-resolving simulations using the System for Atmospheric Modeling (SAM) Khairoutdinov and Randall, 2003. Following C. Muller, 2013, a low-level shear profile is imposed (see also Rotunno, Klemp, and Weisman, 1988; Houze, 1977; Robe and Emanuel, 2001). The wind profile is linear, with a basal velocity of $8.5\,\mathrm{m\,s^{-1}}$ and a shear depth of 1 km. This value is deliberately suboptimal relative to the shear magnitude ($\sim 10\,\mathrm{m\,s^{-1}}$) that maximizes convective organization, in order to enhance sensitivity to internal variability (Abramian, C. Muller, and Risi, 2022). In this regime, the balance between shear and cold-pool dynamics is more unstable, and variations in cold-pool properties exert a substantial control on convective development.

The model domain spans $128\,\mathrm{km} \times 128\,\mathrm{km}$ horizontally and extends from the surface to 17 km in the vertical, with enhanced resolution in the boundary layer. A sponge layer is applied near the model top to absorb upward-propagating gravity waves. Microphysical processes are represented using a single-moment scheme. The simulation is first integrated for 30 days to reach radiative–convective equilibrium and then continued for an additional 27 days for analysis. Model output is saved at 2-min intervals, yielding approximately 2×10^4 time steps, and includes both two- and three-dimensional fields. Under the imposed shear, convection self-organizes into squall-line structures oriented perpendicular to the shear direction, as illustrated in Fig. 1. The system alternates between quiescent phases and episodes of strong organization. Despite identical large-scale forcing, pronounced variability emerges in the structure, intensity, and persistence of the organized convective systems.

2.2 Isentropic mass flux analysis and preprocessing

Vertical transport associated with moist convection is inherently three-dimensional, intermittent, and multiscale. When diagnosed directly in physical space, the vertical mass flux reflects the superposition of contributions from convective updrafts and downdrafts, gravity waves, compensating subsidence, and

turbulent fluctuations. This combination produces a noisy, high-dimensional field, making it difficult to isolate the collective transport by convection or to compare it consistently across realizations.

A physically grounded reduction of this complexity is provided by isentropic analysis Pauluis and Mrowiec, 2013. Rather than organizing vertical motion by spatial location, this approach conditions transport on thermodynamic variables that are approximately conserved following moist adiabatic motion Chen, Leung, et al., 2022. In the moist atmosphere, we approximate isentropic surfaces using the moist static energy variable h. Projecting the vertical mass flux onto h therefore reorganizes transport according to the energetic content of moving air parcels, rather than their instantaneous spatial position.

We define the moist static energy and the associated vertical isentropic mass flux as

$$h(x,y,z,t) = c_p\, T(x,y,z,t) + g\, z + L_v\, q_v(x,y,z,t), \tag{1}$$

$$M(h,z,t) = \int_0^{N_x} \int_0^{N_y} \rho(z,t)\, w(x,y,z,t)\, \delta\Big(\hat{h}(x,y,z,t) - h\Big)\, dx\, dy, \tag{2}$$

where $h(x,y,z,t)$ denotes the moist static energy, c_p is the specific heat capacity of air at constant pressure, $T(x,y,z,t)$ is the absolute temperature, g is the gravitational acceleration, z is height, L_v is the latent heat of vaporization, and $q_v(x,y,z,t)$ is the water vapor mixing ratio. The quantity $M(h,z,t)$ denotes the vertical isentropic mass flux, defined as the horizontal integral of the vertical mass flux $\rho(z,t)\, w(x,y,z,t)$ projected onto moist static energy levels using the Dirac delta function $\delta(\cdot)$. Here $\rho(z,t)$ is the air density, $w(x,y,z,t)$ is the vertical velocity, $\hat{h}(x,y,z,t)$ denotes the local moist static energy field, and N_x and N_y are the horizontal domain dimensions.

This thermodynamic conditioning performs two key functions. First, it reduces dimensionality by collapsing horizontal spatial variability into a compact (h,z) phase space that retains the dominant structure of vertical transport. Second, it separates dynamically distinct processes that overlap in physical space but differ in thermodynamic character. Convective updrafts preferentially transport high h air upward, entraining and evaporatively cooled downdrafts populate lower h levels, while gravity-wave motions—characterized by weak net thermodynamic transport—are confined to low h regions of the phase space(Chen, Pauluis, and F. Zhang, 2018; Chen, Pauluis, Leung, et al., 2020).

Within this isentropic representation, vertical transport organizes into coherent and interpretable patterns (Mrowiec, Pauluis, and F. Zhang, 2016). Sustained positive mass flux at high h extending through the troposphere is characteristic of deep convection, while shallow convection appears as weaker transport rapidly diluted by entrainment. Precipitating systems exhibit enhanced downdraft signatures and broadened h distributions, and cold-pool dynamics manifest as near-surface, low-h mass flux associated with evaporative cooling.

The learning dataset is constructed from the time series of isentropic mass flux fields spanning 27 days with 2-minute temporal resolution. Moist static energy h is discretized into 48 bins ranging from 320 to 350 kJ kg^{-1}, and the vertical coordinate corresponds to the first 48 model levels, spanning altitudes from approximately 25 m to 16 km. The resulting raw dataset has dimensions ($n_t \approx 2\times10^4$, $n_h = 48$, $n_z = 48$). For each spatial pixel, the temporal mean is removed to suppress the stationary background signal and isolate transient variability. A spatial mask is then applied to retain only pixels that exhibit nonzero values at least once over the full time series, thereby discarding invariant regions of the domain; this

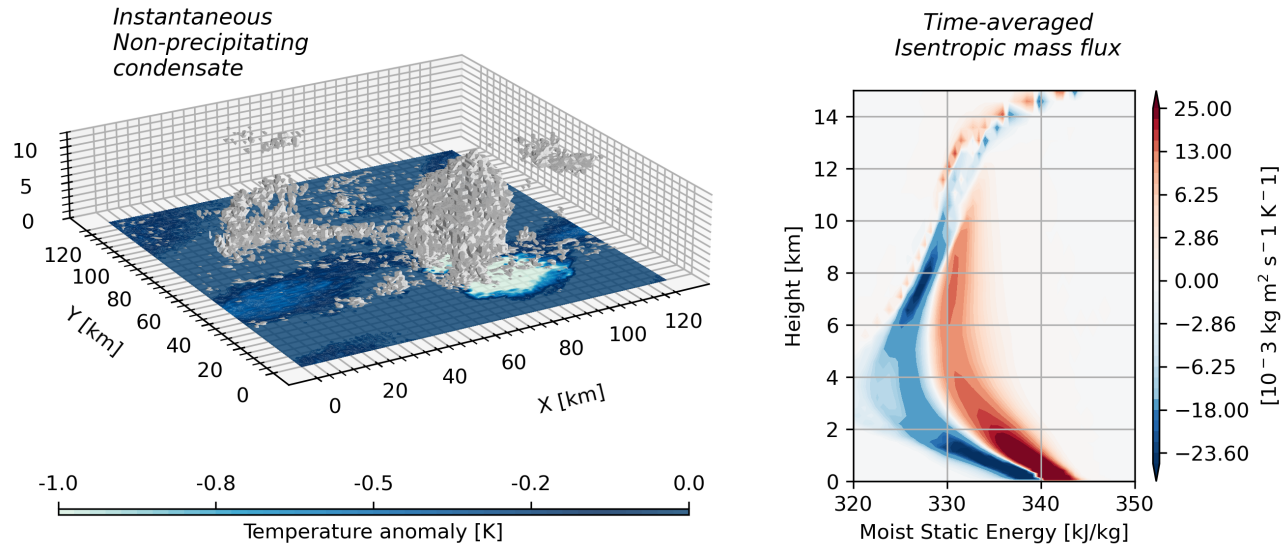


Figure 1: A. Snapshot of the isovolume of non precipitating condensate (QN) at the levels 0f 0.02 kg/kg in volume and the surface temperature anomaly, the white zone corresponding to a cold pool. B. Time average over 27 days with 2min resolution of the isentropic mass flux.

is particularly important at higher altitudes, where the admissible range of h progressively shrinks. The retained fields are subsequently reshaped into vectors. To enhance weak signals while preserving amplitude ordering, a signed logarithmic transform,

$$\operatorname{sign}(x)\,\log(1+|x|),$$

is applied to the masked values. The resulting mean-centered, masked, log-transformed vectors constitute the input representation used to train the model.

Given the time-ordered sequence of processed state vectors $\{X_t\}$, each training sample consists of a pair $(X_t, X_{t+\Delta t})$ separated by a fixed temporal offset $\Delta t = 5$. The full dataset is randomly permuted and split into a training set (90%) and a validation set (9.9%), with the remaining fraction excluded from optimization. During training, samples are drawn in mini-batches of size 256 and randomly shuffled at each epoch to decorrelate temporal ordering within batches, whereas the validation set is iterated deterministically.

2.3 Reduced-order stochastic model

The function of the designed model is not only to reduce the dimensionality of isentropic mass-flux diagrams, but to construct a dynamical, probabilistic representation of convective evolution. Specifically, we seek a reduced representation that (i) encodes the essential physical structure of convection, (ii) enables a meaningful partitioning of convective configurations into discrete states, and (iii) allows us to predict how convection is likely to evolve in time within this reduced space.

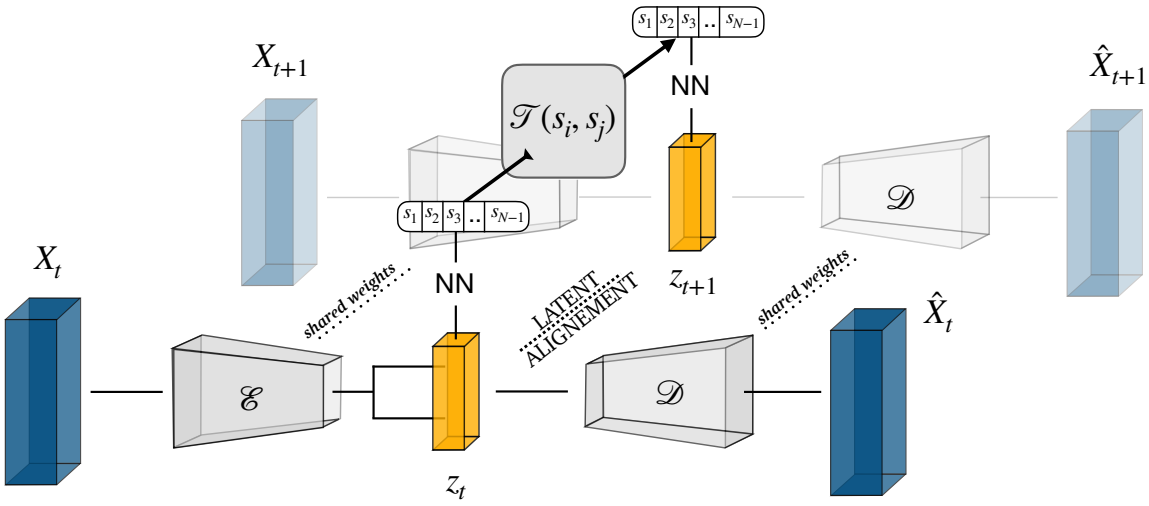


Figure 2: Model description. The model combines a variational autoencoder (VAE) with a Markovian state assignment and transition framework. An encoder maps the input data to a latent representation z_t, which is reconstructed by a decoder. A neural network assigns to each latent vector z_t a probability distribution over a finite set of discrete states. Temporal evolution in latent space is modeled by a linear transition operator T, which predicts the state distribution at time $t+1$ from the state distribution at time t. During training, consecutive latent vectors (z_t, z_{t+1}) are jointly considered: the VAE is trained using a reconstruction loss, while the transition operator is trained by comparing the predicted state distribution $T\,s(z_t)$ with the state distribution inferred from z_{t+1}.

To this end, we introduce a reduced-order stochastic model that combines a variational autoencoder (VAE), a data-driven latent-state representation, and a discrete-time Markov transition model (Fig. 2). Rather than treating dimensionality reduction and temporal modeling as separate steps, the model is trained jointly so that the learned latent representation is both physically meaningful and dynamically predictive.

2.3.1 Variational autoencoder backbone

Each isentropic mass-flux diagram at time t, denoted X_t, is encoded into a d-dimensional latent representation by a variational autoencoder. The encoder outputs a diagonal Gaussian posterior over the latent variable $z \in \mathbb{R}^d$,

$$q(z \mid X_t) = \mathcal{N}\Big(\mu(X_t),\, \mathrm{diag}(\sigma^2(X_t))\Big)\,,$$

where $\mu(X_t)$ and $\sigma(X_t)$ are the mean and standard-deviation vectors produced by the encoder network. A latent sample $z \sim q(z|X_t)$ (drawn via the reparameterization trick) is decoded to produce a reconstruction $\hat{X}_t$, and reconstruction fidelity is enforced through the per-pixel summed squared error $\|X_t - \hat{X}_t\|^2$. The learned posterior is regularized toward an isotropic standard-Gaussian prior $p(z) = \mathcal{N}(0, I)$ through the closed-form Kullback–Leibler divergence $D_{\mathrm{KL}}(q\|p) = \frac{1}{2}\sum_{i=1}^{d}(\mu_i^2 + \sigma_i^2 - \log\sigma_i^2 - 1)$. This regularization yields a smooth latent space in which nearby points correspond to similar convective configurations, enabling meaningful interpolation and supporting a stable, probabilistic description of convective evolution.

2.3.2 Latent Space Partitioning

The continuous latent representation is mapped onto a finite set of $K = 7$ discrete convective states through a small neural network that takes z_t as input and outputs a probability vector via a softmax head, $s_t = \mathrm{softmax}(g(z_t)) \in \Delta^{K-1}$, with $s_t(k) \geq 0$ and $\sum_k s_t(k) = 1$. The number of clusters K is set empirically (cluster-count sensitivity in Fig. S4). This mapping can be interpreted as a data-driven clustering mechanism, in which close latent representations are grouped into dynamically meaningful regimes. Importantly, because the clustering network is trained jointly with the encoder, it does not merely partition a fixed latent space but actively reshapes the latent geometry to favour cluster-consistent representations. The assignment is probabilistic rather than deterministic, allowing for uncertainty and overlap between states, which is essential in transitional or weakly organized configurations.

To prevent state collapse—whereby most samples are assigned to a small subset of states—we regularize the state assignment using a contrastive objective that encodes prior physical knowledge about the input. Specifically, we introduce a contrastive loss based on the structural similarity of isentropic mass-flux anomaly fields. Each sample is discretized into three regions using fixed thresholds $\{-1.5, 1.5\}$, corresponding to negative anomalies, near-neutral values, and positive anomalies. From these regions, binary masks are constructed and compared using a macro-averaged Dice coefficient, which measures the spatial overlap of positive and negative anomaly structures while intentionally discarding information about anomaly amplitude or fine-scale intensity. For randomly sampled pairs (i, j) within a batch, the Dice similarity $\mathrm{Dice}(i, j)$ defines a notion of physical proximity. Pairs with high structural similarity are encouraged to share the same latent state, while structurally dissimilar pairs are encouraged to occupy different states. This is implemented through a soft contrastive loss that penalizes disagreement for similar pairs (cohesion) and agreement for dissimilar pairs (separation), with a smooth transition between the two regimes. An additional mid-range penalty discourages uncertain assignments when the similarity is ambiguous. We denote this term as $\mathcal{L}_{\mathrm{Dice}}$ and adopt the convention that it is a *similarity score* (larger when state assignments match the Dice-based structural similarity); accordingly, it enters the total loss with a negative sign so that maximizing similarity *minimizes* the total objective. Full definition in Supplementary Information.

2.3.3 Markov transition dynamics

We assume that, at the temporal resolution of the simulations, the evolution of convection can be approximated as a first-order Markov process in state space. The transition probabilities are encoded in a transition matrix T, defined by

$$T_{ij} = P(s_{t+\Delta t} = j \mid s_t = i),$$

with rows that are non-negative and sum to unity (row-stochastic). The data are sampled every $\Delta t_{\mathrm{raw}} = 2\,\mathrm{min}$, but we evaluate transitions at $\Delta t = 5\,\Delta t_{\mathrm{raw}} = 10\,\mathrm{min}$ (motivated in Section 3.3). This matrix captures the persistence of convective regimes, preferred transitions, and rare events such as the onset or decay of organized convection. During training, the model receives frame pairs $(X_t, X_{t+\Delta t})$, encodes them into $(z_t, z_{t+\Delta t})$, and assigns states $(s_t, s_{t+\Delta t})$; T is then fit so that $\hat{s}_{t+\Delta t} = T\, s_t$ matches $s_{t+\Delta t}$.

2.3.4 Joint optimization

The VAE, state-assignment network, and transition matrix are trained jointly by minimizing

$$\mathcal{L} = \underbrace{\|X - \hat{X}\|^2}_{\text{reconstruction}} + \underbrace{\tfrac{1}{2}\sum_{i=1}^{d}\left(\mu_i^2 + \sigma_i^2 - \log\sigma_i^2 - 1\right)}_{\text{KL to } \mathcal{N}(0,I)} - \underbrace{\mathcal{L}_{\text{Dice}}}_{\text{contrastive}} \underbrace{- \sum_{i,j} s_t(i)\, s_{t+1}(j)\, \log T_{ij}}_{\text{transition log-likelihood}} .$$

Each term is weighted by a non-negative coefficient (β, λ_{Dice}, γ_{MC}); values, together with encoder/decoder architectures, are reported in the Supplementary Information.

These four terms exert competing pressures on the latent geometry. The KL term pulls samples toward an isotropic Gaussian, while the Dice and transition terms favor a multi-modal structure in which samples cluster into K regions separated by both structural similarity and temporal predictability. The resulting equilibrium is a latent space whose envelope remains approximately Gaussian but whose interior carries $K = 7$ resolvable clusters (Fig. 3).

Optimization uses AdamW (learning rate 5×10^{-5}, batch size 256) for 512 epochs with gradient-norm clipping at 1.0; the best model is selected by minimum validation loss. Loss decompositions (Fig. S1) and reconstruction diagnostics (Fig. S2) are provided in the Supplementary Information.

3 Results

3.1 Physical interpretation of the latent space

The model compresses time series of isentropic mass-flux fields—used here as a physically grounded representation of the convective state—into a time series of latent vectors $\mathbf{z}(t) \in \mathbb{R}^8$, with each time step assigned to one of seven discrete convective states.

Latent-space structure revealed by PCA. A natural question is whether the latent space carries any physical meaning. There is no guarantee that it should: no physical constraint was imposed during training—the VAE is regularized only toward an isotropic Gaussian, and the seven discrete states are assigned without reference to intensity, organization, or any other diagnostic. To probe the structure of the latent space, we apply PCA to the latent vectors $\mathbf{z}(t)$ and examine how the discrete clusters are arranged along its dominant modes of variability. The first two components explain 22% and 17% of the variance, respectively (Fig. 3); a three-dimensional projection (Supplementary Information, Fig. S5) confirms that clusters overlapping in 2D are separated along higher-order components.

Emergent axes: convective life cycle and organization. The physical interpretation of the dominant latent directions is obtained by relating the principal components to independent diagnostics of convection (Fig. 4). The first principal component is strongly correlated with domain-mean precipitation, indicating that it primarily captures convective intensity and progression along the convective life cycle, from weak or suppressed states to vigorous, precipitating convection.

In contrast, the second principal component shows little correlation with precipitation but is strongly anticorrelated with a measure of cloud spatial organization derived from the variance of precipitable

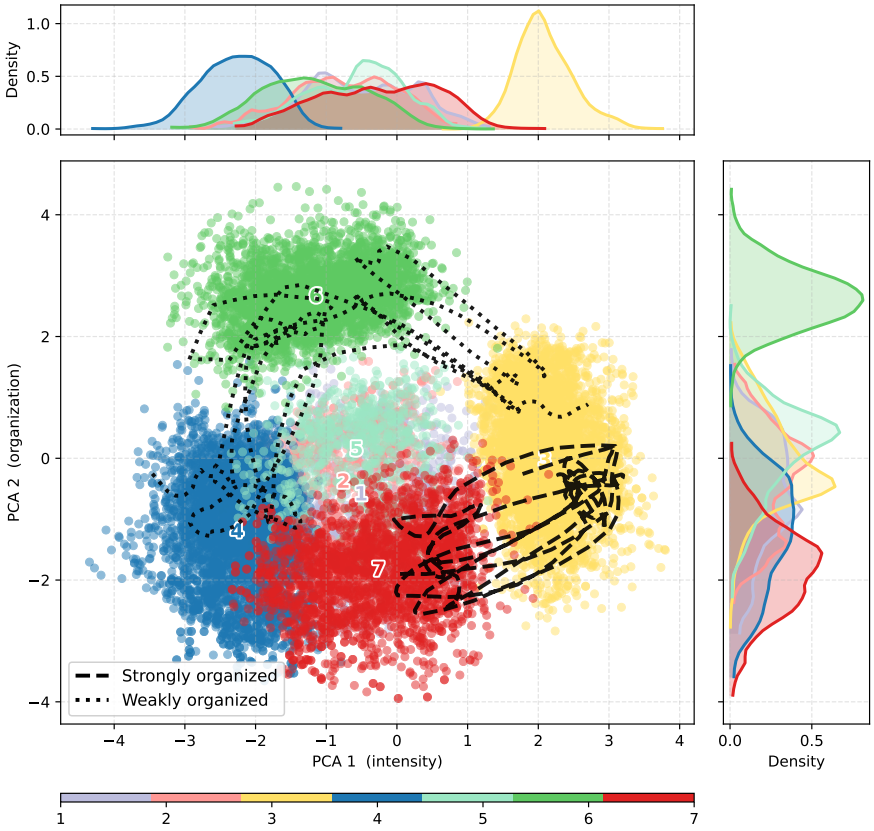


Figure 3: Latent-space organization. PCA projection of the latent space colored by cluster assignment. Each point corresponds to one time step; trajectories illustrate the time evolution over 5 h.

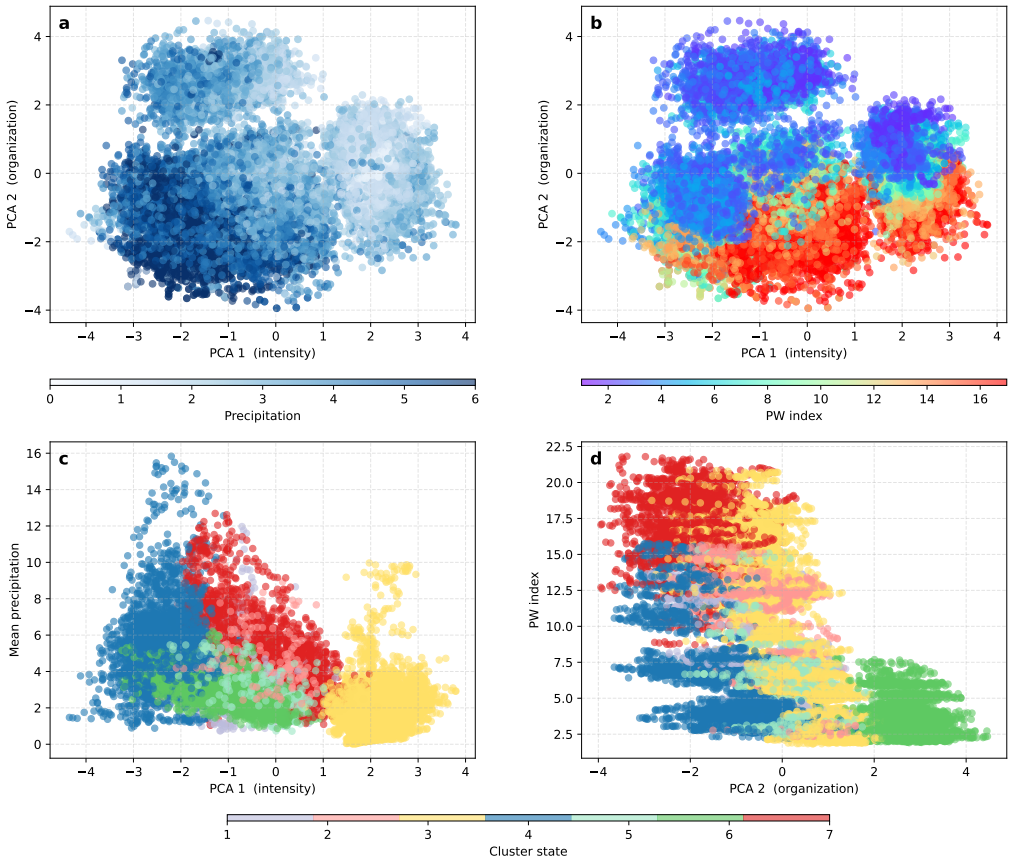


Figure 4: Physical interpretation of latent variability. PCA projection colored by (A) mean precipitation and (B) organization index. (C–D) Joint distributions of the first two principal components with the corresponding diagnostics.

water. Variability along this axis therefore reflects differences in convective organization rather than intensity. The near-orthogonality of the two components indicates that convective intensity and spatial organization emerge as largely independent modes of variability in the learned representation.

Dynamical and thermodynamic signatures of latent states. If the latent clusters correspond to physically meaningful convective regimes, they should be associated with systematic differences in vertical mass transport and thermodynamic structure. Representative isentropic mass-flux fields for each cluster are shown in Supplementary Information (Fig. S7).

Let $M(h, z, t)$ denote the isentropic mass flux at time t, height z, and moist static energy h. For each cluster c, we define the cluster-mean vertically integrated positive and negative mass fluxes,

$$M_c^{\pm}(z) = \left\langle \int_h M(h, z, t)\, \mathbf{1}\{\pm M > 0\} \right\rangle_{t \in c}, \tag{3}$$

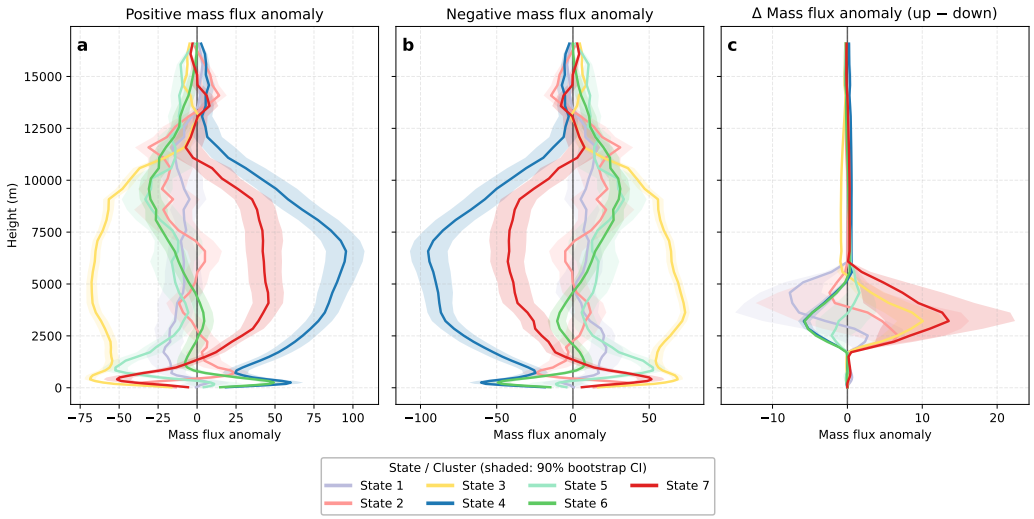


Figure 5: Cluster-averaged vertical mass-flux anomalies. Positive, negative, and net anomalies relative to the simulation-wide mean.

and anomalies relative to the simulation-wide mean profiles $\overline{M}^{\pm}(z)$,

$$\Delta_c^{\pm}(z) = M_c^{\pm}(z) - \overline{M}^{\pm}(z), \qquad \Delta_c^{\text{net}}(z) = \Delta_c^{+}(z) - \Delta_c^{-}(z). \tag{4}$$

The resulting vertical profiles (Fig. 5) reveal clear dynamical distinctions between clusters. States separated primarily along the intensity axis differ mainly in the magnitude of both upward and downward mass flux throughout the troposphere. Cluster 3 exhibits weak updrafts and downdrafts, consistent with shallow or suppressed convection, whereas Cluster 4 shows enhanced updraft–downdraft coupling characteristic of intense, precipitation-driven convection. Cluster 6 displays moderate, broadly distributed up- and downdrafts with limited vertical coherence, consistent with weakly-organized, popcorn-like convection. Cluster 7 displays the strongest net upward mass transport over much of the troposphere, consistent with sustained deep convection.

Thermodynamic differences associated with convective organization are evident in the distribution of mass-flux anomalies across moist static energy (Fig. 6). Organized states, notably Cluster 7, exhibit enhanced positive mass flux at high moist static energy, indicative of reduced entrainment and enhanced buoyancy within coherent convective structures. Less organized states show anomalies concentrated at lower energy levels, consistent with stronger entrainment and more transient convection.

Together, these results show that the VAE latent space organizes convective variability along two emergent and physically interpretable axes—intensity and organization—and that the resulting discrete latent states correspond to dynamically and thermodynamically distinct convective regimes. Table 1 consolidates the per-cluster numerical and physical characterization.

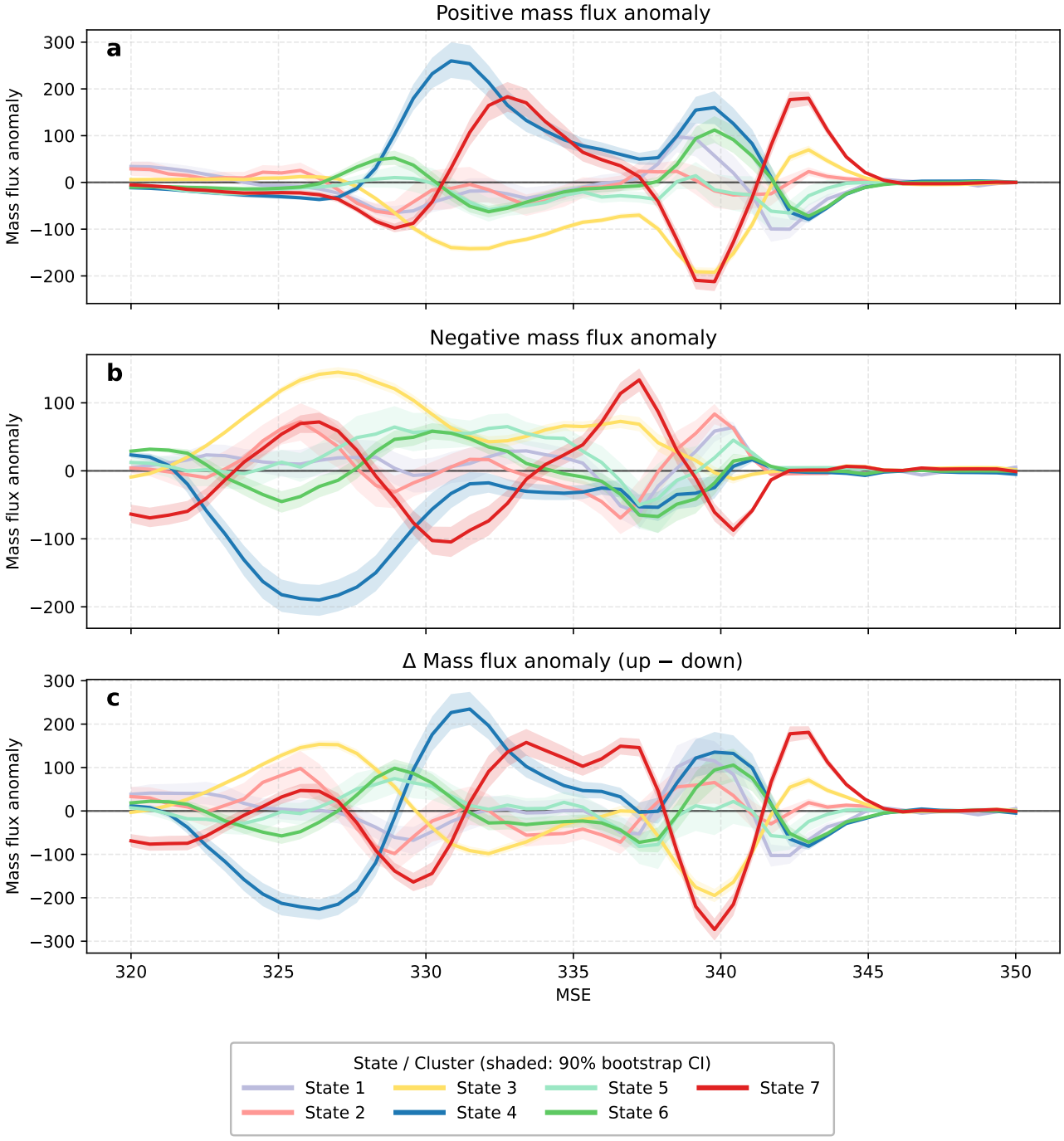


Figure 6: Cluster-averaged moist static energy–resolved mass-flux anomalies.

State	Name	π (%)	Physical signature
1	*Transient*	4.6	rare, short-lived; faint mass-flux anomalies; no clear life-cycle role
2	*Bridge*	3.5	weak mid-level updrafts; preferred gateway between organized and disorganized regimes
3	*Quiescent*	36.4	suppressed/shallow convection; weak up- and down-drafts; dry background atmosphere
4	*Mature*	19.1	strongly coupled updraft–downdraft column; heavy precipitation; cold-pool generation
5	*Outlier*	4.0	rare, short-lived; weakly distinguishable mass-flux signature
6	*Popcorn*	19.7	scattered, weakly-organized shallow-to-congestus convection; moderate precipitation
7	*Organized deep*	12.7	deep, sustained updraft cores with extensive stratiform downdrafts; squall-line-like organization

Table 1: Summary of the seven latent convective states. π_i is the stationary probability of state i (the right-eigenvector of $\mathbf{T}$ associated with the unit eigenvalue, satisfying $\mathbf{T}\boldsymbol{\pi} = \boldsymbol{\pi}$; Eq. 6). The chain's dynamical relaxation is governed by the six subdominant *eigenmodes* of $\mathbf{T}$ (not by the seven clusters individually), with physical time scales $\tau_k = -\Delta t / \log|\lambda_k|$ where $\Delta t = 10\,\text{min}$ is the chain step and the index k labels the eigenmode (not a cluster): $\tau_2 = 144\,\text{min}$ (2.4 h), $\tau_3 = 63\,\text{min}$ (1.0 h), $\tau_4 = 53\,\text{min}$ (0.9 h), $\tau_5 = 37\,\text{min}$ (0.6 h), $\tau_6 = \tau_7 = 26\,\text{min}$ (0.4 h). The autocorrelation of any cluster i is a sum of these exponentials weighted by the cluster's amplitude in each eigenvector $\mathbf{v}_k$ (Eq. 10), with the leading mode τ_2 setting the longest decay (Fig. 8). Clusters with strong projection on $\mathbf{v}_2$ (notably 3, 4, 6, 7) inherit a ∼2-hour persistence; clusters that project mainly on faster modes (5 on $\mathbf{v}_3$) decay in ∼1 h.

3.2 Markov transition matrix, eigenmodes, and regimes

The second output of the model is a Markov transition matrix $\mathbf{T}$, whose elements

$$T_{ij} = \mathbb{P}\left(X_{t+\Delta t} = j \mid X_t = i\right) \tag{5}$$

denote the probability of transitioning from state i to state j. The resulting transition matrix is shown in Fig. 7.

The matrix $\mathbf{T}$ is strongly diagonally dominated, indicating that the system tends to persist within a given state over several time steps. Moreover, the matrix is sparse, with only a limited number of preferred transitions between states, suggesting constrained pathways in state space.

To further characterize the dynamics, we perform a spectral decomposition of $\mathbf{T}$. Because $\mathbf{T}$ is stochastic, it admits a unit eigenvalue $\lambda_1 = 1$. The corresponding right eigenvector $\boldsymbol{\pi}$ satisfies $\mathbf{T}\boldsymbol{\pi} = \boldsymbol{\pi}$ and defines the stationary distribution of the Markov chain. We obtain

$$\boldsymbol{\pi} = \begin{bmatrix} 4.6\% & 3.5\% & 36.4\% & 19.1\% & 4.0\% & 19.7\% & 12.7\% \end{bmatrix}. \tag{6}$$

The stationary distribution is dominated by Cluster 3 (36.4%), followed by Clusters 6 (19.7%), 4 (19.1%), and 7 (12.7%). Clusters 1 (4.6%), 2 (3.5%), and 5 (4.0%) are rarely occupied.

We next examine the subdominant eigenvalues and eigenvectors of $\mathbf{T}$. All non-stationary eigenvalues have imaginary parts at the level of numerical noise ($|\text{Im}(\lambda_k)| < 10^{-3}$); the spectrum is effectively real. This indicates that the dominant dynamics cannot be described as simple linear oscillations in state space.

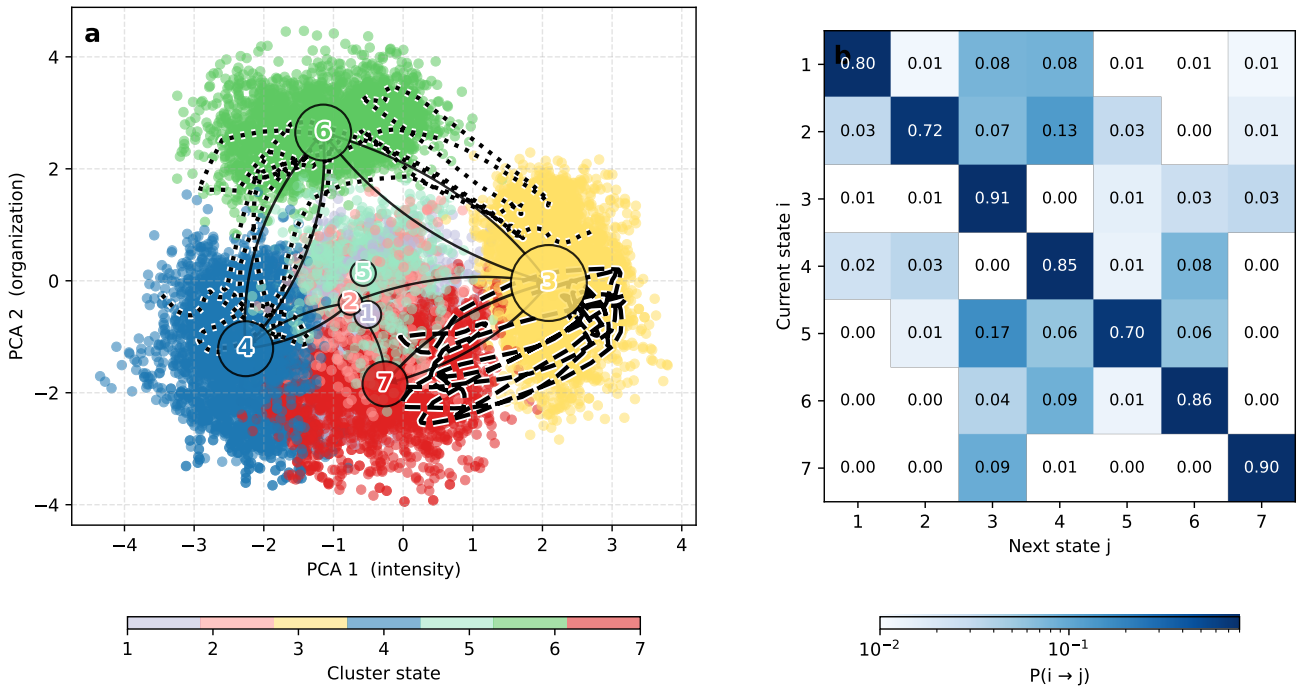


Figure 7: Markov state dynamics in latent space. (A) Latent space visualization with the inferred Markov chain overlaid; example trajectories along the two dominant pathways are drawn on top (the corresponding time series are shown in Supplementary Information, Fig. S6). The size of each state node is proportional to its asymptotic (stationary) probability, and arrows indicate transition probabilities between states. For clarity, transitions involving States 1 and 5 are omitted: both have negligible stationary occupancy ($\pi < 5\%$) and very weak amplitudes in the leading eigenmode (Table 1). **(B)** Transition matrix describing the dynamics of convection under a given shear. The dominance of the diagonal indicates strong state persistence, with the system remaining in a given state for extended periods. Analysis of the transition structure reveals preferred pathways, including oscillations among States 3–6–4 associated with weakly organized convection, and transitions between States 3 and 7 associated with strongly organized convection. The matrix further indicates that transitions from weakly to strongly organized convection preferentially occur via the intermediate State 2.

The leading nontrivial eigenvalue of the transition matrix is

$$\lambda_2 \approx 0.93, \tag{7}$$

indicating a highly persistent dynamical mode. The corresponding eigenvector

$$\mathbf{v}_2 = \begin{bmatrix} -0.11 & -0.19 & +0.30 & -0.49 & -0.02 & -0.44 & +0.65 \end{bmatrix} \tag{8}$$

(the unit-norm right eigenvector; its overall sign is arbitrary) describes deviations from the stationary distribution. Positive (negative) components of $\mathbf{v}_2$ indicate enhanced (suppressed) occupancy of the corresponding states relative to $\boldsymbol{\pi}$.

This eigenmode separates the dynamics into two contrasting regimes. In its positive phase, Clusters 3 and 7 are preferentially occupied while Clusters 4 and 6 are suppressed. In the negative phase the pattern reverses: Clusters 4 and 6 are enhanced at the expense of Clusters 3 and 7. Clusters 1 and 2 carry smaller but non-negligible negative amplitudes (≈ -0.11 and -0.19), aligning weakly with the negative-phase regime, while Cluster 5 carries essentially no signal in this mode ($|\text{amplitude}| < 0.05$). These two regimes are also apparent directly in the structure of the transition matrix $\mathbf{T}$.

Physically, the two phases of $\mathbf{v}_2$ correspond to distinct modes of convective evolution. The positive phase (Clusters 3 and 7 enhanced) represents a directly organized pathway, in which suppressed conditions transition into sustained deep convection. The negative phase (Clusters 4 and 6 enhanced) describes the alternative route, in which convection passes through intense precipitation-driven and unorganized, popcorn-like phases before dissipating. Time series of individual realizations (Supplementary Information, Fig. S6) confirm that system trajectories cluster along these two pathways, consistent with the structures inferred from the transition matrix and its leading eigenmodes.

The eigenvalue spectrum just identified makes a quantitative prediction: for a true Markov chain, the autocorrelation of every state's occupancy must be a sum of exponentials with the decay times $\tau_k = -\Delta t / \log|\lambda_k|$. In the next subsection we test this prediction directly against the empirical autocorrelations, identifying which states deviate from Markovian behavior and how those deviations reveal internal feedbacks such as cold-pool memory.

3.3 Markovianity assessment and internal feedbacks

We finally test whether the seven-state chain provides a self-contained description of convective evolution, by asking whether the state at time t alone is sufficient to predict the state at $t + \tau$. Deviations from this Markov property would indicate slow internal processes—moisture memory, cold pools, mesoscale organization—that the partition does not resolve. The question is long-standing: classical convective parameterizations assume instantaneous quasi-equilibrium adjustment (G. J. Zhang, 2003; Bechtold et al., 2014), while more recent schemes introduce explicit memory variables to capture lagged responses (Piriou et al., 2007; Gerard et al., 2009; Mapes and Neale, 2011; Hwong et al., 2023).

Indicator autocorrelation. Defining the occupancy indicator $I_i(t) = \mathbf{1}[X_t = i]$ and its normalized autocorrelation

$$C_i(\tau) = \frac{\mathbb{E}[(I_i(t) - \pi_i)(I_i(t+\tau) - \pi_i)]}{\mathbb{E}[(I_i(t) - \pi_i)^2]}, \tag{9}$$

a discrete-time Markov chain with transition matrix $\mathbf{T}$ and stationary distribution $\boldsymbol{\pi}$ admits the closed form

$$C_i(\tau) \;=\; \frac{T_{ii}^{\tau} - \pi_i}{1 - \pi_i} \;=\; \sum_{\alpha=2}^{K} A_{\alpha}^{(i)}\, \lambda_{\alpha}^{\tau}, \tag{10}$$

a superposition of exponentially decaying modes set by the eigenvalues λ_α of $\mathbf{T}$, with state-dependent amplitudes $A_\alpha^{(i)} = (\mathbf{v}_\alpha)_i(\mathbf{w}_\alpha)_i/(1-\pi_i)$, where $\mathbf{v}_\alpha$ and $\mathbf{w}_\alpha$ are the right and left eigenvectors of $\mathbf{T}$ (derivation in SI). Departures from this prediction quantify residual memory not encoded by the partition.

Memory beyond Markov in the convection-active states. Most states' empirical autocorrelations track Eq. 10 (Fig. 8a). States 6 and 7 decay measurably more slowly than their own Markov prediction. This distinction matters: a state can be *sustained* (long dwell times) and still *Markovian* if its diagonal T_{ii} alone explains the persistence; States 6 and 7 are sustained *beyond* what T_{ii} predicts, indicating that an internal degree of freedom not represented in the partition is keeping them alive.

Cold pools partly close the gap. Cold pools are a natural candidate carrier of this missing memory (Tompkins, 2001; Houze, 1977; Rotunno, Klemp, and Weisman, 1988). The hypothesis is testable: if cold pools carry the memory, then knowing the cold-pool field at the starting time should make the subsequent State-7 dynamics more Markovian—faster autocorrelation decay, closer to the Markov prediction. To test this we introduce an indicator that picks out State-7 timesteps with a high cold pool,

$$J_7^{\star}(t) \;=\; \mathbf{1}[X_t = 7 \text{ and } \mathrm{CP}(t) > \mathrm{CP}_{\star}]\,, \tag{11}$$

where $\mathrm{CP}_\star$ is the 75th percentile of the domain-integrated cold-pool intensity. The cold-pool-conditioned State-7 survival is then the expected State-7 occupancy a lag τ later, averaged over the timesteps selected by $J_7^\star$:

$$S_7^{(\mathrm{CP})}(\tau) \;=\; \mathbb{E}\big[\, I_7(t+\tau) \,\big|\, J_7^{\star}(t) = 1 \,\big]\,. \tag{12}$$

We then rescale to autocorrelation form via $C_7^{(\mathrm{CP})}(\tau) = (S_7^{(\mathrm{CP})}(\tau) - \pi_7)/(1-\pi_7)$ so that the result can be plotted on the same axis as panel (a). The cold-pool condition is enforced *only at the starting time*; after t, the trajectory is free to wander in CP. Enforcing $J_7^\star = 1$ throughout the lag window would mechanically truncate the survival whenever CP dropped below $\mathrm{CP}_\star$—an artifact of the conditioning, not of the dynamics (SI compares the two formulations). The conditioned curve (Fig. 8b, dashed) sits between the unconditional empirical curve (solid) and the Markov reference (dotted), closing roughly 20–30% of the gap at lags of 100–200 min. Knowing the cold-pool field at the starting time therefore makes the subsequent State-7 dynamics more Markovian—but not fully Markovian.

Cold pools are part—but not all—of the hidden memory. The unclosed portion of the gap implicates other slow internal variables: mesoscale moisture pooling (Mapes and Neale, 2011), cloud-cluster geometry (Hwong et al., 2023), and free-tropospheric humidity all evolve on the relevant

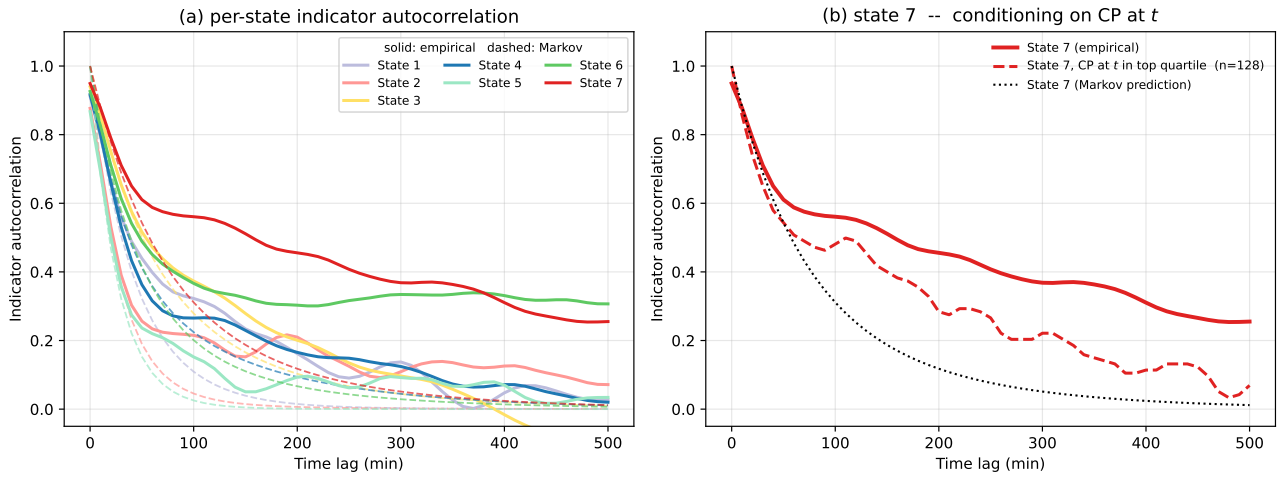


Figure 8: (a) Per-state indicator autocorrelations (solid) compared with the Markov prediction for the same state (dashed, same colour). For most states the empirical curve tracks the prediction closely, indicating that the seven-state Markov chain captures their dynamics well. States 6 and 7 decay noticeably more slowly than their own Markov prediction, retaining residual memory not encoded by the seven discrete states alone. **(b)** Cold-pool conditioning recovers part of the State-7 memory. Three curves on the same autocorrelation axis as panel (a): (i) *solid red* is the empirical State-7 autocorrelation (same as in panel (a)); (ii) *dotted black* is the State-7 Markov prediction—what the autocorrelation *would* look like if the seven-state chain were fully Markovian (identical to the red dashed curve of panel (a)); (iii) *dashed red* is the same State-7 autocorrelation, but now restricted to starting times at which the cold-pool intensity CP(t) exceeds its 75th-percentile threshold CP$_\star$ (the cold-pool condition is applied only at the starting time; the trajectory is free thereafter). The dashed red curve sits between the solid red and the dotted black—knowing the cold-pool field at the starting time shifts the dynamics closer to Markov, revealing part of the hidden memory, but does not close the gap entirely.

time scales. Two caveats apply: CP(t) collapses spatial structure into a scalar, and stratifying by CP(t) inherits some lifecycle-stage dependence; SI sensitivity tests (per-cold-pool diagnostic and entry-conditional survival anchored at State-7 entries) are consistent with the closure reported here. A definitive identification of the dominant carrier would require re-training the encoder with explicit cold-pool inputs, which we leave to future work.

4 Conclusion and perspective

We developed a data-driven framework that treats convective organization dynamically: an encoder maps isentropic mass-flux profiles into a low-dimensional latent space, the latent space is partitioned into discrete convective states, and transitions between states are modelled as a Markov chain. Applied to cloud-resolving squall-line simulations under fixed vertical wind shear, this construction addressed three questions: *(i)* what mass-flux states a given forcing makes accessible; *(ii)* whether transitions between them can be approximated by a Markov chain, and how organized and weakly organized pathways differ; *(iii)* where internal feedbacks introduce effective memory beyond the discrete state.

Convective variability lives on a low-dimensional manifold captured by eight latent dimensions and seven discrete states that recover the canonical phases of the convective life cycle. The chain reveals two competing routes through this state space: a direct ascent into sustained deep convection, and a longer drift through weakly organized phases. States 6 and 7 retain residual memory beyond their Markov

prediction. Conditioning on cold-pool intensity at the starting time closes 20–30% of this gap at lags of 100–200 min.

The framework's distinctive capability is that departures from Markovianity *diagnose* what the discrete partition is missing, identifying the slow internal variables that a closed dynamical description of convection must include.

Within a given forcing, the Markovianity diagnostic can be sharpened by conditioning on richer descriptors of internal state than the domain-integrated cold-pool intensity used here. Because the isentropic mass-flux decomposition retains the downdraft branch of the flow, the representation already exposes the slower time scale on which evaporative downdrafts—and the cold pools they generate—develop relative to the convective updraft; conditioning explicitly on cold-pool age, gust-front strength, and other lifecycle markers would target this development lag directly. Per-cold-pool quantities capture the spatial structure that a scalar index averages out and could be aggregated by cold-pool shape, alignment, or proximity to neighbours; mesoscale moisture pooling, free-tropospheric humidity, and cluster-geometry measures each isolate a further candidate slow variable that the latent state does not currently resolve. Each addition that further closes the empirical-versus-Markov gap identifies another slow variable a memory-aware parameterization must track; the full attribution of the residual State-7 memory is the most direct extension of the present work.

The simulations analysed here span a single shear configuration. The most immediate extension is to repeat the construction across forcing regimes—weak shear (popcorn convection), moderate shear (mixed regimes), strong shear (squall lines)—and compare the resulting transition matrices. The expected shifts are concrete and dynamically interpretable: under stronger shear, the diagonal probabilities T_{ii} of the organized states should increase, the direct pathway from initiation into organized deep convection should dominate over the indirect route through weakly organized phases, and the corresponding eigenmodal time scales should lengthen. Mapping the transition matrix as a function of forcing therefore turns the organization itself into a measurable dynamical observable.

5 Data and Code Availability

The simulations analyzed in this study consist of idealized squall-line simulations performed with the System for Atmospheric Modeling (SAM). The dataset spans 27 days of simulation at 2-minute temporal resolution and includes one-dimensional, two-dimensional, and three-dimensional model outputs. The data are available via Globus (link to be provided).

All code used in this study is openly available at: https://github.com/sososquirrel/troposwhere. The repository includes the implementation of the model as well as the scripts required to reproduce all figures presented in the paper.

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7 Author Contributions

S.A. designed the research, performed the simulations, developed the methods and code, and wrote the manuscript. O.P. contributed to the methodological development, contributed to the code, and participated in the analysis and discussions. P.G. contributed to the interpretation of the results and to writing the manuscript.

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